Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold

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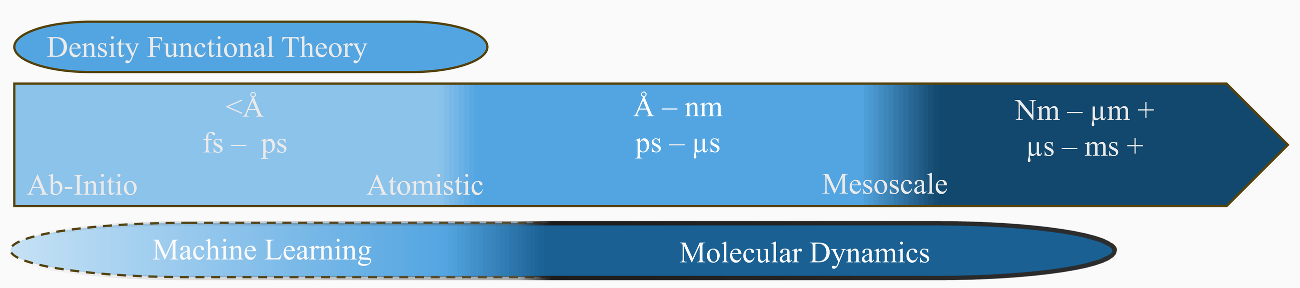
Abstract

Classical molecular dynamics is a powerful method of theoretically describing the properties of a material but is often inconsistent with quantum molecular dynamics. Quantum molecular dynamics while more accurate are much more computationally intensive and not feasible for systems of more than 1000 atoms. Machine learning is used to bridge the gap between quantum accuracy and the scalability of classical interatomic potentials. Here, we report on progress made testing the performance of a quantum-accurate classical interatomic potential for gold against state-of-the-art classical potentials in describing experimentally verified properties of gold using LAMMPS (Large Scale Atomic/Molecular Massively Parallel Simulator) Preliminary results show close agreement with quantum molecular dynamics simulations, while being computationally less expensive. Future work includes extending this study to simulate rare events in a large, such as the herringbone reconstruction of the Au(111) surface, that are not accurately described by current classical potentials.

Types of Simulations



- First Principles simulation methods such as Density Functional Theory give quantum accurate data about a system
- Molecular Dynamics is a versatile numerical method that is able to simulate much larger system sizes
- We aim to use machine learning to extend the use of molecular dynamics to have the accuracy of first principles methods at a lower computational cost, allowing for larger simulations.

Molecular Dynamics

- Molecular Dynamics is solving a small representative system numerically.
- We approximate the atoms as 'particles' and solve their associated equations of motion.
- In the microcanonical ensemble, defined as keeping the number of particles, volume and energy of a system fixed, we use Newton's Second Law.

$$\sum F = ma$$

- The Canonical Ensemble is defined as keeping the number of particles, volume, and temperature fixed in the system. In this case we use the Langevin equation, which introduces two terms to account for the thermal fluctuations. These are used as a 'thermostat' for our system.

$$m\frac{d^2\vec{r}_i}{dt^2} = -\lambda\frac{d\vec{r}_i}{dt} + \eta(t) - \nabla_i U(\vec{r}_1,\dots,\vec{r}_n)$$

- There is a common potential energy term between these equations of motion. The methods of molecular dynamics used are based on how this term is defined.

Acknowledgements

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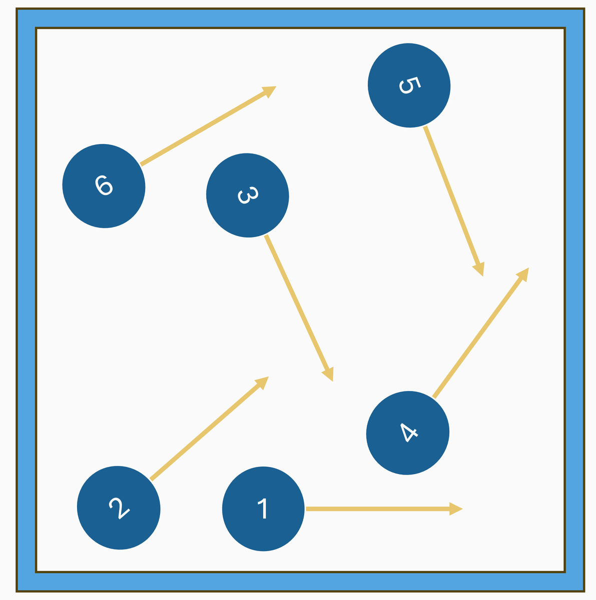
Radial Distribution Function and Stress Strain Curve of Physical Gold:

- (D) Warren, B. E. Radial distribution functions from electron diffraction. J. Appl. Crystallogr. 32, 1012–1015 (1999). https://doi.org/10.1107/S0021889899006603
- (A) Suzuki, T., A. Vinogradov, and S. Hashimoto. "Strength Enhancement and Deformation Behavior of Gold after Equal-Channel Angular Pressing." Materials Transactions. doi:10.2320/matertrans.45.2200.

Example System



Solving Equations of Motion for 6 Particles

$$\sum F = ma$$

Position vector of i[th] atom

$$\vec{r}_i = x_i\hat{x} + y_i\hat{y} + z_i\hat{z}$$

Acceleration as 2[nd] derivative of position

$$a_i = \frac{d^2\vec{r}_i}{dt^2}$$

Force is found through the negative gradient of potential energy

$$F_i = -\nabla_i U(\vec{r}_1,\vec{r}_2,\vec{r}_3,\vec{r}_4,\vec{r}_5,\vec{r}_6)$$

From this we construct 18 2[nd] order differential equations.

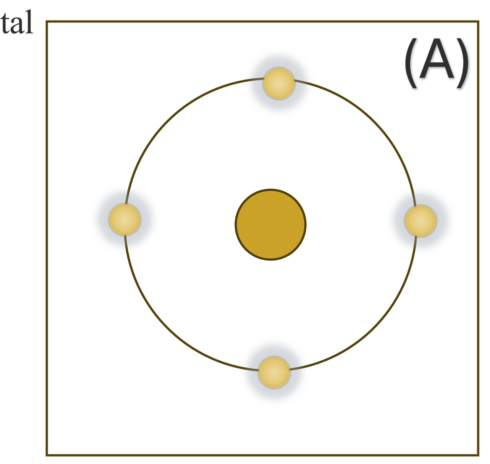
$$m_i\frac{d^2\vec{r}_i}{dt^2} = -\nabla_i U(\vec{r}_1,\dots,\vec{r}_6)$$

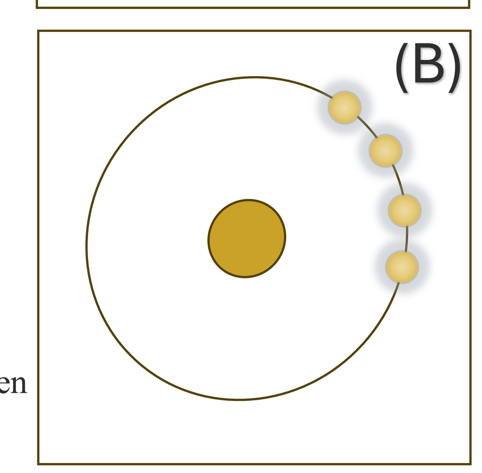
Types of Potentials

- Embedded Atom Method (EAM) potentials define the total energy as a pairwise interaction potential energy and an energy required to place an electron into a spherically averaged electron density.

$$E_{tot} = \sum_i f(\bar{\rho}_i) + \frac{1}{2}\sum_{\substack{i,j \\ j\neq 1}} U_{i,j}(r_{i,j}) + E_{error}$$

- The force coefficients of this term are empirically determined.
- Because of the spherical symmetry, EAM is not able to discern between the configurations of (A) and (B).



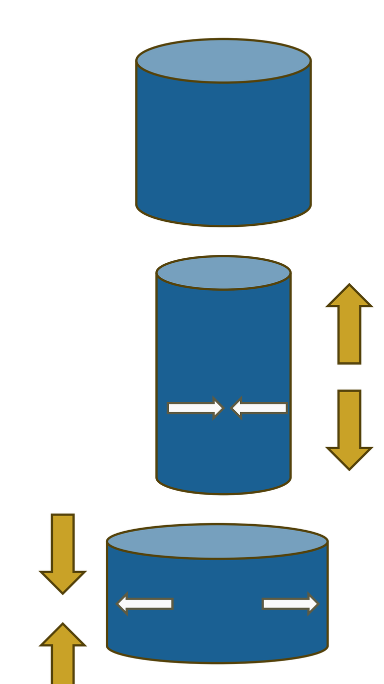


- Machine learning potentials such as Atomic Cluster Expansion (ACE) define the total energy as the sum of basis function contributions.

$$E_{tot} = \sum_i U_i = \sum_i \sum_\alpha c_\alpha B_\alpha^{(i)}$$

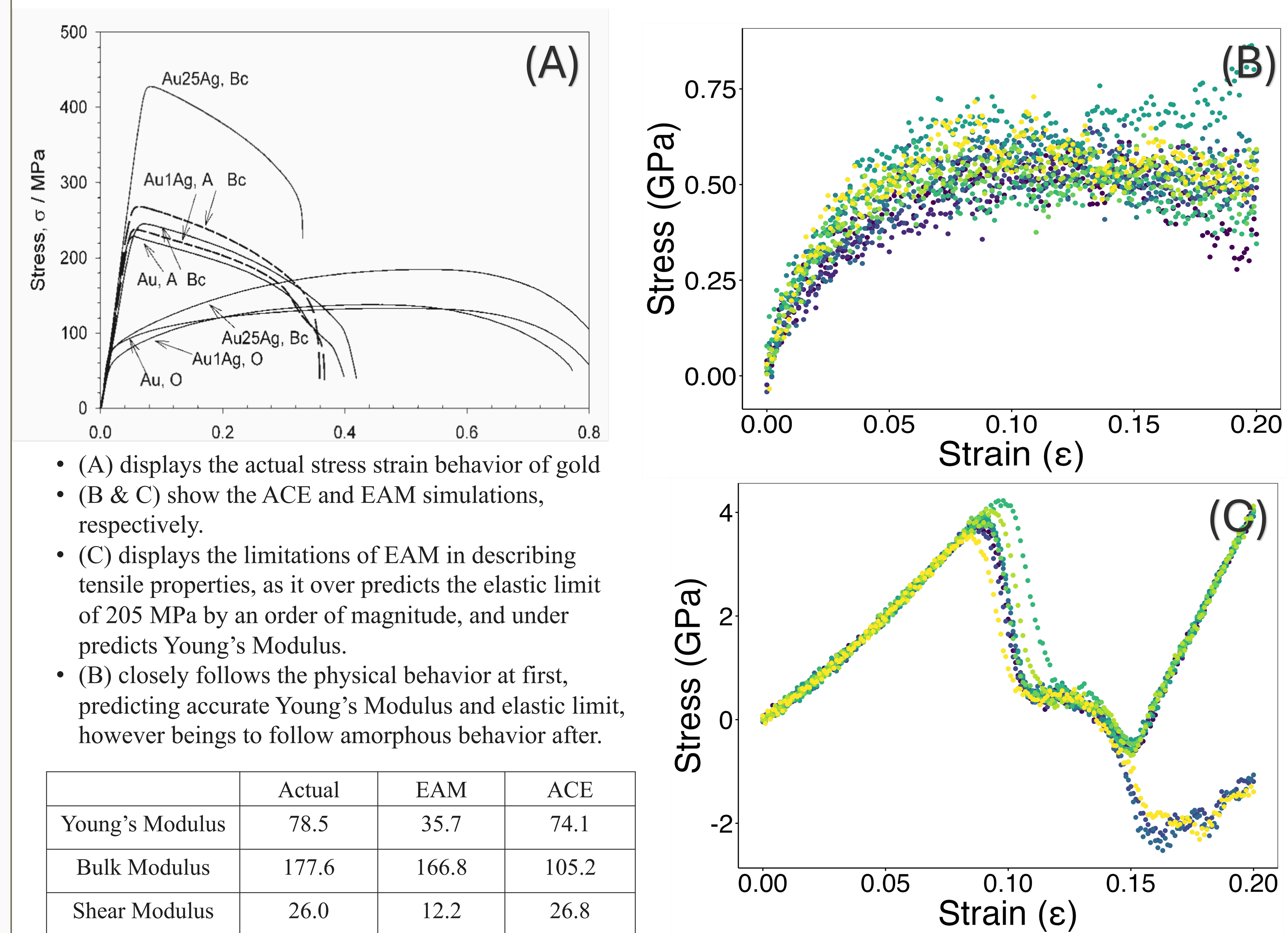
- ACE uses spherical harmonics of the atom as basis functions to encode angular dependence allowing it to accurately describe the potential energy difference between (A) and (B).
- Coefficients are fit to these basis functions using a neural network.

Mechanical Properties



- Stress-strain tests are common physical tests to determine mechanical properties of materials. By stretching and compressing a material we can gain insights into its material properties. We will compare our simulation methods with physically verified data to benchmark their effectiveness.
- LAMMPS is used to perform our molecular dynamics simulations. For each potential, the simulation box is altered to stretch or compress the bulk gold volumetrically for bulk simulations and uniaxially for stress-strain curves.
- Elastic constants are extracted using the components of the pressure tensor and the lengths of the simulation box.

Results



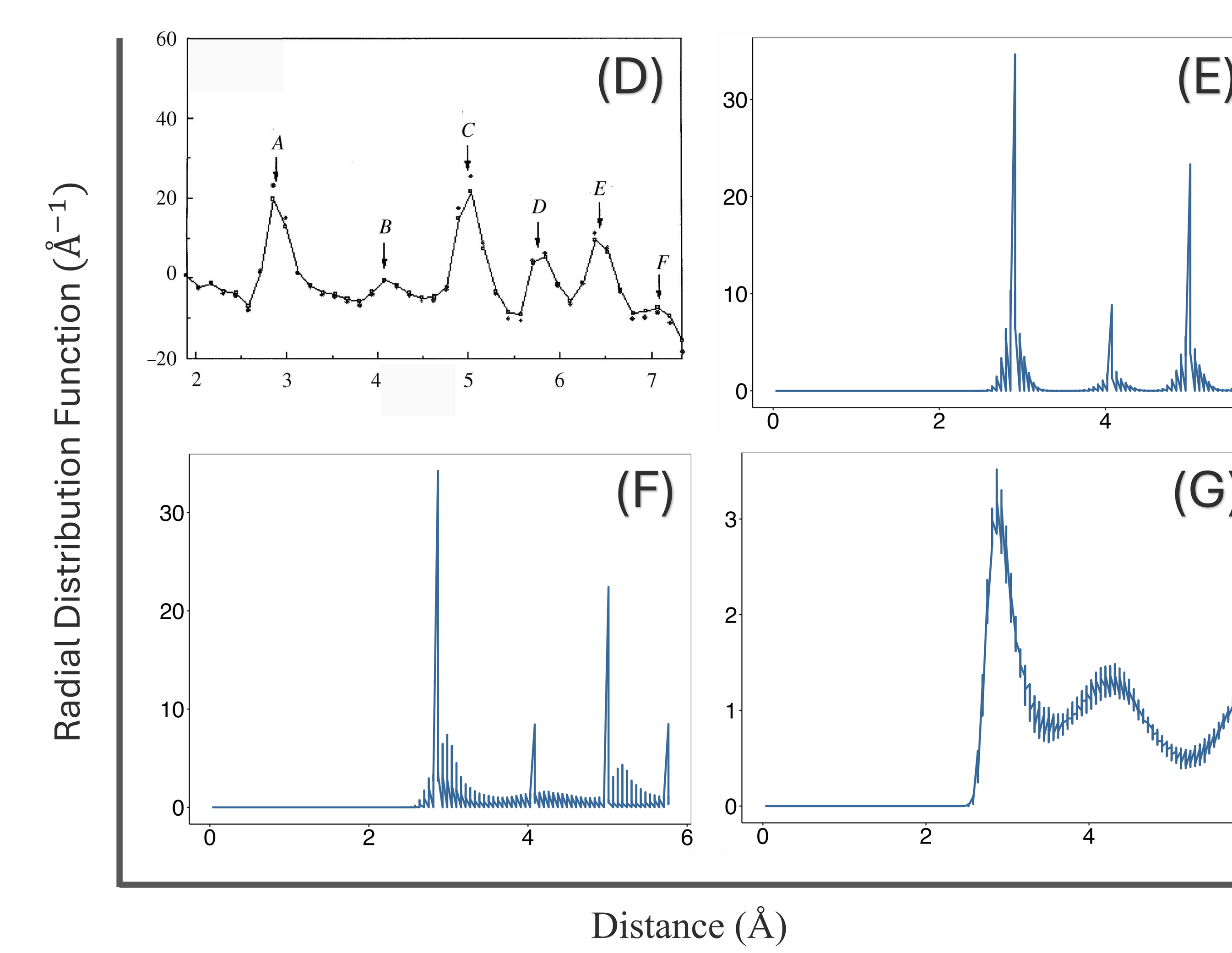
- (A) displays the actual stress strain behavior of gold
- (B & C) show the ACE and EAM simulations, respectively.
- (C) displays the limitations of EAM in describing tensile properties, as it over predicts the elastic limit of 205 MPa by an order of magnitude, and under predicts Young's Modulus.
- (B) closely follows the physical behavior at first, predicting accurate Young's Modulus and elastic limit, however beings to follow amorphous behavior after.

	Actual	EAM	ACE
Young's Modulus	78.5	35.7	74.1
Bulk Modulus	177.6	166.8	105.2
Shear Modulus	26.0	12.2	26.8
Poisson's Ratio	0.44	0.46	0.38

- (D) – Physical Gold RDF
- (E) – EAM
- (F) – ACE at 0K
- (G) – ACE at 300K

- Sharp peaks correspond to crystalline structures, wide peaks with lower magnitudes describe amorphous (liquid) behavior, a straight line would describe an ideal gas.

- (E) and (F) show close agreement with the magnitudes and locations of peaks in the RDF distribution.
- (G) shows that when simulated at room temperature, the ACE model predicts a liquid.



Conclusion

- The tested ACE potential incorrectly predicted liquid behavior at room temperature of gold.
- Training data was fit without any temperature dependence, causing the neural network to not know how to predict the behavior at temperature, to fix this we will supplement training data.
- The model however shows potential as an improvement over EAM. The incorrect training data only means that our coefficients are fitted incorrectly, and thus when corrected may vastly improve.
- Despite the errors at large temperatures, the model closely predicts tensile constants at low temperature where EAM fails to properly predict, showing its potential.